Xinwei Qiang

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Education

University of California San Diego — Ph.D. in Computer Science Sep 2026 – Present
Advisor: Prof. Yufei Ding

Shanghai Jiao Tong University — B.E. in Computer Science Sep 2022 – Jun 2026
John Hopcroft Class, an honors program in computer science.

Research Interests

I build GPU systems for large-model training and inference: attention kernels and their scheduling, compilers that generate multi-GPU kernels overlapping computation with communication, and serving systems for LLMs and diffusion transformers. I also apply these techniques to zero-knowledge proof computation.

Publications

[PPoPP '27] Plectra: Scheduling Parallelism for Diffusion Transformer.
Xinwei Qiang, Yifan Hu, Shixuan Sun, Jing Yang, Han Zhao, Chen Chen, Yu Feng, Jingwen Leng, Minyi Guo

[NeurIPS '26] AsymHP: Load-Balanced Sparse Attention for Video Diffusion Transformers.
Xinwei Qiang*, Yue Guan*, Ruihan Zhu, Mihir Jagtap, Zaifeng Pan, Zhongkai Yu, Chang Chen, Zhengding Hu, Yufei Ding, Adnan Aziz

[ASPLOS '27] Impeller: Scalable and Efficient Zero-Knowledge Proof Computation Framework via Graph Compilation.
Liukun Yu*, **Xinwei Qiang***, Shixuan Sun, Chen Chen, Jingwen Leng, Jin Dong

[SC '26] FluxZK: Scalable and Efficient Zero-Knowledge Proof Computation via GPU Acceleration.
Xinwei Qiang*, Liukun Yu*, Xiyu Wang, Zhengyi Li, Shixuan Sun, Jingwen Leng, Chen Chen, Jiaping Gui, Zhenzhe Zheng, Jin Dong, Minyi Guo

[ICML '26] TritonGym: A Benchmark for Agentic LLM Workflows in Triton GPU Code Generation.
Yue Guan*, Yichen Lin*, Xu Zhao, Jianzhu Yao, **Xinwei Qiang**, Zhongkai Yu, Pramod Viswanath, Yufei Ding, Adnan Aziz

[OSDI '26] Syncopate: Efficient Multi-GPU AI Kernels via Automatic Chunk-Centric Compute-Communication Overlap.
Xinwei Qiang*, Yue Guan*, Zhengding Hu, Keren Zhou, Yufei Ding, Adnan Aziz

[ICLR '26] DASH: Deterministic Attention Scheduling for High-throughput Reproducible LLM Training.
Xinwei Qiang, Hongmin Chen, Shixuan Sun, Jingwen Leng, Xin Liu, Minyi Guo

[SOSP '25] Mercury: Unlocking Multi-GPU Operator Optimization for LLMs via Remote Memory Scheduling.
Yue Guan, **Xinwei Qiang**, Zaifeng Pan, Daniels Johnson, Yuanwei Fang, Keren Zhou, Yuke Wang, Wanlu Li, Yufei Ding, Adnan Aziz

* Equal contribution.

Preprints

[arXiv '26] Beyond Parallel Blindness: Information Floors and Model Gaps in Block Drafting.
Xinwei Qiang, Xiang Fang, Chang Chen, Zaifeng Pan, Yue Guan, Yufei Ding

Research Appointments

Research Assistant, EPCC Lab, Shanghai Jiao Tong University Jul 2024 – Jun 2026
Co-advised by Profs. Jingwen Leng and Shixuan Sun

Research Intern, Picasso Lab, University of California San Diego Jul 2025 – Dec 2025
Advisor: Prof. Yufei Ding

Service

Reviewer: ICLR 2027

Technical Skills

Languages: C/C++, CUDA, Python, Rust, Triton
Frameworks: PyTorch, SGLang, vLLM-Omni